

MARS RECONNAISSANCE ORBITER NAVIGATION STRATEGY FOR THE COMET SIDING SPRING ENCOUNTER

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Comet Siding Spring encountered Mars on October 19, 2014 at a distance of about 140,500 km – the nearest comet flyby of a planet in recorded history. Mars Reconnaissance Orbiter (MRO) was able to detect the comet, gather science data, and capture images of the comet as it approached Mars. To help protect MRO from the incoming comet particles, two propulsive maneuvers were performed to position the spacecraft behind Mars at the arrival time of the expected peak particle fluency. This paper documents the strategy that the MRO Navigation Team executed to mitigate risk from the comet particles while allowing scientific observations of the comet flyby.

INTRODUCTION

Mars Reconnaissance Orbiter (MRO) is just completing 10 years of operations since its launch on August 12, 2005. After the Mars Orbit Insertion (MOI) on March 10, 2006, the MRO spacecraft concluded the Primary Science Phase (PSP) in November 2008, the Extended Science Phase (ESP) in September 2010, and two extended missions (EM1 & EM2) in September 2012 and 2014, respectively. MRO is currently operating in its third extended mission (EM3) scheduled through September 2016. Following MOI, the MRO Navigation Team has been providing trajectories and other navigation products to the project per the navigation requirements.^{1,2} The models and filter strategy used currently in navigation operations are similar to that shown in Reference 3. MRO has achieved over 42,000 orbits and all the instruments on-board are functioning well and continuing to provide valuable science data. The MRO Mission is a key component of NASA's Mars Exploration Program (MEP), providing both fundamental science, and MEP support, including relay, landing site selection & certification, and atmospheric information for other MEP missions. MRO has high-resolution imagery, context imagery, hyperspectral imaging spectrometry, atmospheric profiling, weather camera observations, ground penetrating radar observations, and gravity experiments. Additionally, using its UHF antenna MRO supported the Entry, Descent and Landing (EDL) phases for two martian landers: Phoenix in May 2008³ and Curiosity in August 2012.⁴ The MRO Team is currently planning to support the EDL sequence of NASA's InSight Mission in September 2016.⁵

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MRO also continues to relay data to Earth from the currently operating Mars rovers, Opportunity and Curiosity.

On October 19, 2014, Comet Siding Spring, also known as C/2013 A1, had a close flyby of planet Mars. To prepare for this celestial visitor, MRO and other Mars orbiters performed evasive maneuvers and thereby avoided the peak fluency of comet particles. This paper documents the MRO navigation strategy to mitigate the risk posed by the comet particle grains while allowing the spacecraft to image the comet and acquire science data during portions of the comet flyby of Mars.

MRO PRIMARY SCIENCE ORBIT

The Primary Science Orbit (PSO) for MRO operations is a $252 \text{ km} \times 317 \text{ km}$ altitude, sun-synchronous orbit with the periapsis frozen over the south pole and the ascending node at 3:00 PM. The mean orbital elements are shown in Table 1. Ideally the orbits are designed to exactly repeat after 4602 orbits in 349 sols (1 sol = 1.0275 days) providing sub-5 km coverage at the equator. The near repeat cycle used for science planning is a 211-orbit cycle (16 sols) that walks about 0.5 deg (32.5 km) in longitude westward from the previous cycle. The orbit maintenance is done based on this near repeat cycle.

Table 1: MRO Mean Orbital Elements

Periapsis Epoch: 02-Jul-2014 12:21:11.492 ET	
Semi-Major Axis (a)	3647.7213 km
Eccentricity (e)	0.0055
Inclination (i)	92.7507°
Argument of Periapsis (ω)	270.3481°
Right Ascension of Node (Ω)	75.4907°
True Anomaly (v)	0.0°
Additional Orbit Information (Apoapsis Epoch: 02-Jul-2014 13:17:25.770 ET)	
Period (T)	111.48 min
Periapsis Altitude (H_p)	251.4051 km
Apoapsis Altitude (H_a)	314.7982 km

COMET SIDING SPRING OVERVIEW

Comet Siding Spring (CSS), originating from the Oort cloud, is a comet discovered on January 3, 2013 by Robert H. McNaught at the Siding Spring Observatory in Australia. It is a long-period comet with its inbound orbit period being several million years while the outbound orbit period is estimated as one million years. It reached its perihelion on October 25, 2014 at a distance of about 1.4 AU. The closest approach to Mars by Comet Siding Spring on October 19, 2014 brought it as close as about 140,500 km,⁶ the nearest passing of a comet to a planet in our solar system's recorded history. Observations of the comet in the spring of 2013 were curtailed due to its close proximity to the Sun as viewed from Earth. However, comet observations in the first half of 2013 and scrutiny of pre-discovery comet images enabled detailed modeling of the comet.^{7,8,9,10} This modeling of the number and size of particles likely to intersect Mars and its orbiting spacecraft showed that the particles able to bridge the gap between the comet's closest approach and Mars would have already been ejected from the comet. The modeling then focused on the times when these particles would reach Mars, as the best risk mitigation strategy for the Mars orbiters was to be on the other side of Mars during the height of particle fluency. MRO took photographs of the comet with its High Resolution Imaging Science Experiment (HiRISE) camera as the comet approached Mars providing the first resolved images of the nucleus of a long-period comet. Hyperspectral imaging was done by the Compact Reconnaissance Imaging Spectrometer for Mars (CRISM) instrument. Possible effects on the Mars environment from the comet's visit were studied by the Mars Climate Sounder (MCS) and the Shallow Radar (SHARAD) antenna. To help protect MRO from the particles that had been ejected from the comet, two propulsive maneuvers were performed in July and September 2014 to position the spacecraft behind Mars at the time of the expected peak fluency of the incoming particles.

SELECTION PROCESS FOR COMET SAFE LOCATION

The particles that had already been ejected from the comet were expected to follow soon after the arrival of the comet. Earlier independent analyses^{7,8,9,10} yielded a consistent arrival time of maximum fluency of these comet particle grains and their direction (right ascension and declination) at Mars. The safe location for MRO from the incoming comet particles was identified as the midpoint of its orbit occulted from the direction of the particles, as shown in Figure 1. In

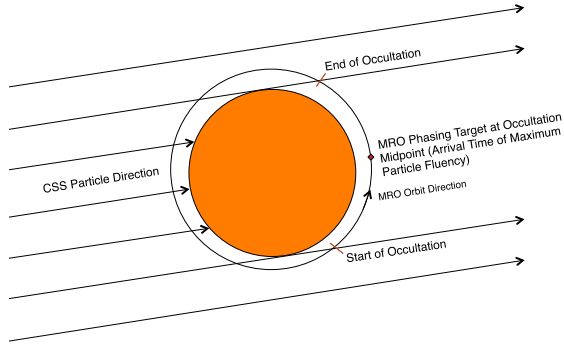


Figure 1: MRO Phasing Target (CSS Safe Location)

order to mitigate any risk to the MRO spacecraft, the Navigation Team was tasked to develop a plan to phase MRO to this location. Specifically, the MRO spacecraft would be phased such that the safe location declination would be reached at the time of the maximum particle fluency, which was determined to be approximately 98 minutes after the comet closest approach to Mars. There was no plan to adjust the right ascension of MRO as it would have required a costly orbit plane change maneuver.

The particle direction and maximum fluency time, based on References 7, 8, 9, and 10, were provided to the Navigation Team by the Mars Program Office via the Comet Encounter Target File (CETF) which contains the particle grain direction's right ascension (RA) and declination in the Mars Mean Equator of J2000 (MME2000) inertial reference frame (Appendix: Comet Encounter Target File). Initially, the MRO Navigation Team verified that the particle grain velocity direction (165.40° RA and 8.50° declination in MME2000) was consistent with the comet velocity direction (165.36° RA and 8.45° declination in MME2000) at the time of closest approach. This verification also served as validation of the CETF.

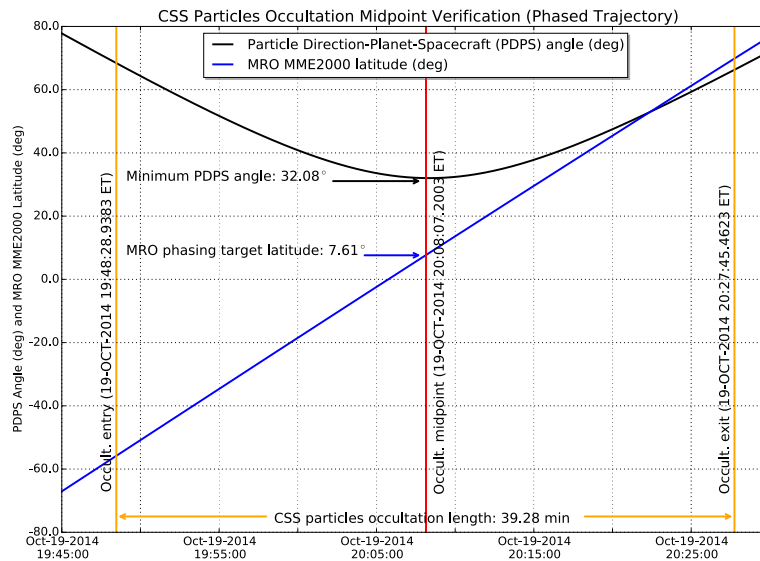


Figure 2: Occultation Midpoint with Phased Trajectory

The midpoint of the occultation duration, which was the identified safe location, lasted for 39.28 minutes - about a third of the MRO orbital period (Figure 2). The occultation entry and exit times

(safe zone) after performing the phasing maneuvers are shown by orange lines in the figure, with the occultation midpoint indicated by the middle red line. The spacecraft location at the midpoint of the occultation duration was verified to be in close proximity to the spacecraft location when the angle between the particle grain velocity direction and the spacecraft's position vector, both vectors relative to the center of Mars, is at a minimum. This angle to minimize will herein be referred to as the Particle Direction-Planet-Spacecraft (PDPS) angle. The agreement between the spacecraft position at the occultation duration midpoint and at the PDPS minimum angle is a result of MRO's nearly circular orbit ($252 \text{ km} \times 317 \text{ km}$). For eccentric orbits, since the spacecraft velocity could be significantly different at various parts of the orbit, the geometric midpoint of the occulted orbit arc rather than occultation duration midpoint would be more appropriate for the safe location. This may be necessary, especially in the case when the occultation duration period is rather short. The safe location was targeted using MRO's latitude at the midpoint of the occultation duration (7.61° in Figure 2). If the geometric midpoint was chosen, the achieved latitude would have been closer to the target declination of 8.5° given in the CETF.

COLLISION AVOIDANCE AT THE COMET SAFE LOCATION

Five spacecraft were operating around Mars at the time of the comet flyby: NASA's MRO, Odyssey, and MAVEN; ESA's Mars Express (MEX); and Mars Orbiter Mission (MOM) from the Indian Space Research Organization (ISRO). Collision risk concerns were raised given that four of the five orbiters would be phased behind Mars at the arrival time of the maximum particle fluency (MEX was instead positioned based on the time of the comet's closest approach to Mars). However, there was no determined risk of collision between any of the spacecraft, due to the different orbits of each spacecraft which did not intersect. The closest approach identified was between MAVEN and MRO, right at the maximum fluency time, but the minimum distance between the two spacecraft was still more than 264 km, abating any collision risk concerns. Figure 3 shows the position of all spacecraft at the time of maximum particle fluency, as seen from a direction opposite to the velocity direction of the comet particles relative to Mars.

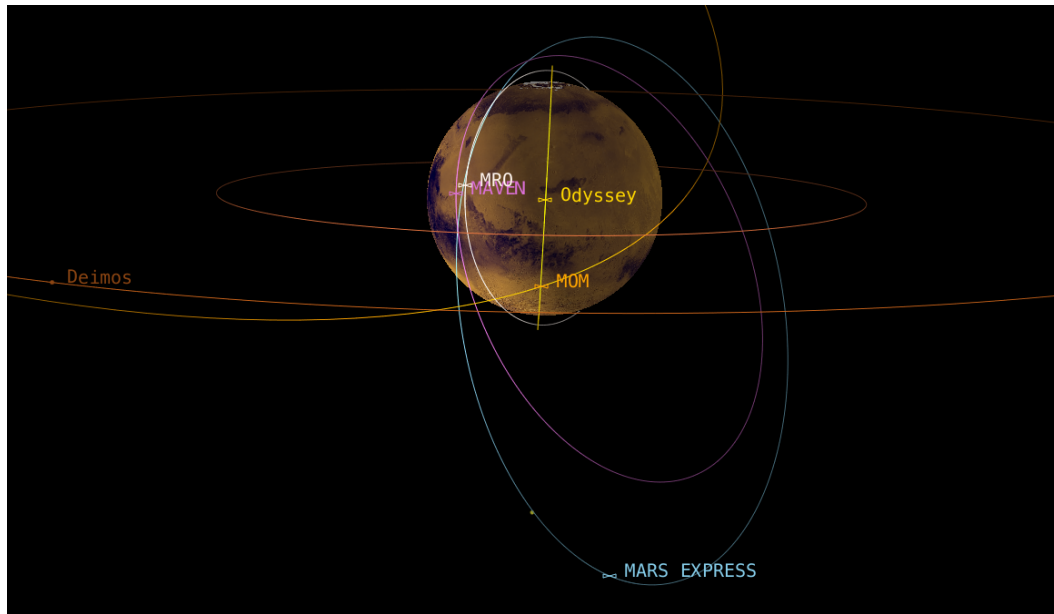


Figure 3: Mars Orbiters at Time of Comet Maximum Particle Fluency

COMET RISK MITIGATION PHASING STRATEGY

The MRO Navigation Team planned to phase MRO to the midpoint of the safe zone behind Mars occulted from the maximum fluency of the arriving comet particles. This was to be done by changing the orbital period and thereby adjusting mostly the spacecraft true anomaly at the target time. An important consideration of this approach was the amount of phasing correction to be made. For MSL EDL support, MRO took the slow down approach as its target location was almost on the other side of the orbit (about 49 minutes of timing offset just prior to the first phasing maneuver) and a pro-velocity maneuver raised the semi-major axis and increased the orbital period.⁴ A benefit of raising the semi-major axis was that MRO experienced less drag due to lower atmospheric density. To phase MRO to the safe location from the arriving comet particles to within ± 2 minutes as given in the CETF (see Figure 14 in Appendix: Comet Encounter Target File), two in-plane Orbit Trim Maneuvers (OTMs) were performed in the anti-velocity direction. The timing offset at the time of OTM-37, the first phasing maneuver, was about 19 minutes late (approximately 1/6 of MRO's ~ 112 minute orbital period). Performing both OTM-37 and OTM-38 to remove the ~ 19 minute phasing offset was the most cost effective approach (i.e., expend less ΔV).

Anticipated Atmospheric Drag and Navigation Timing Uncertainties

The atmospheric density variation is the biggest error source to the MRO navigation accuracy, smaller only to any significantly big maneuver execution error.⁵ As shown in Figure 4, the atmospheric drag was anticipated to be much higher in the time frame leading up to the CSS flyby than at previous times when phasing was performed to support the EDL phases of the Phoenix and MSL missions. Maneuver planning was done anticipating a drag ΔV of 0.3 mm/s per orbit. Consequently, the expected navigation timing uncertainties were significantly larger than at the Phoenix and MSL EDLs (Figure 5). Correcting the phasing only up to the navigation uncertainty level at a given maneuver opportunity would avoid potentially overshooting the target time. Also when planning the phasing strategy it was recognized that the maneuver ΔV should not go below the minimum control capability (2 cm/s). These factors led to the two-maneuver phasing strategy that was adopted for the CSS risk mitigation.

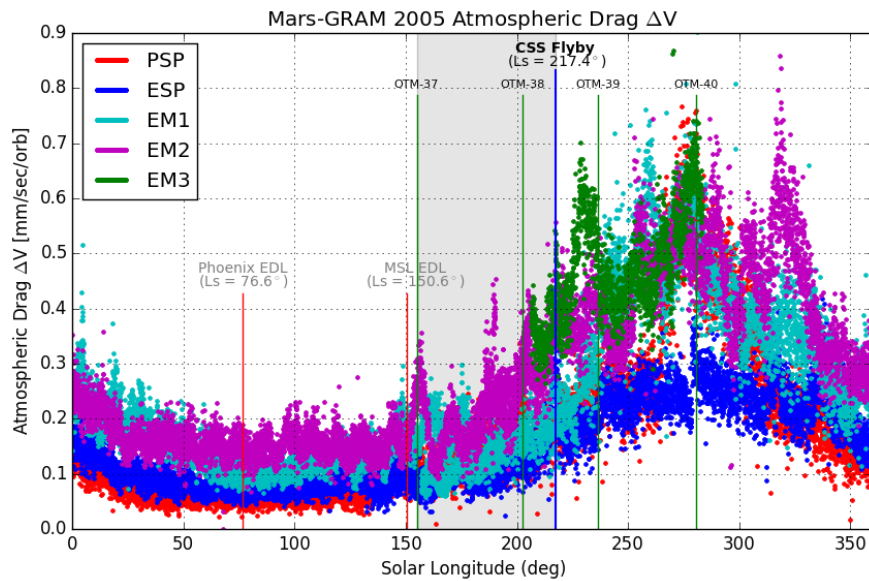


Figure 4: Atmospheric Drag ΔV Experienced by MRO Through January 2015

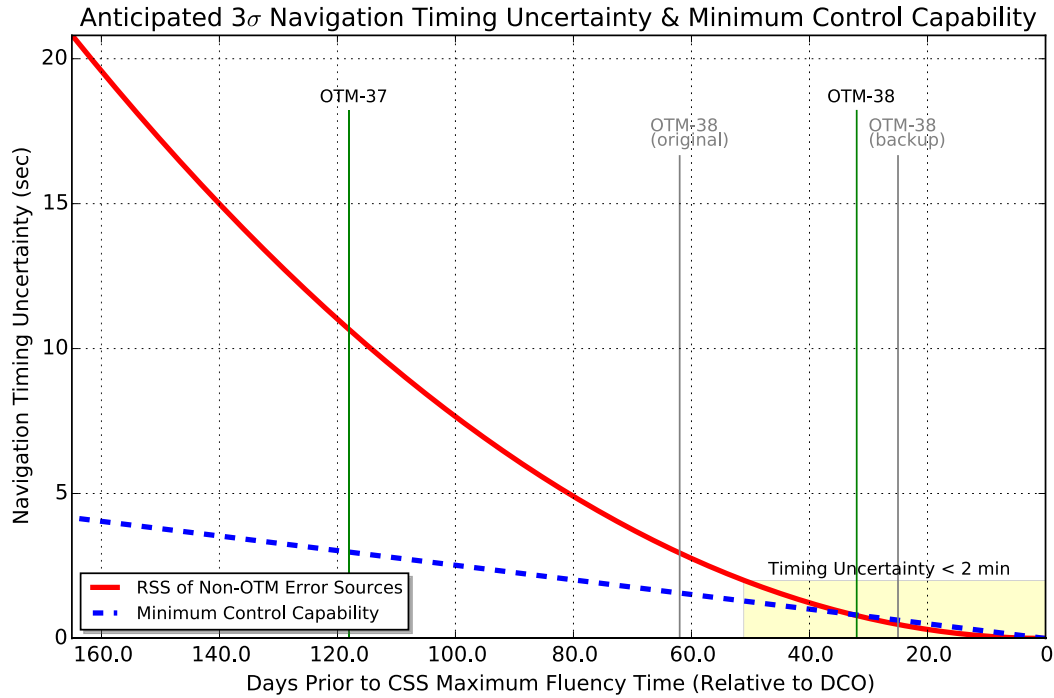


Figure 5: MRO Navigation Timing Uncertainty (3σ) Prior to CSS Closest Approach

Selection of Phasing Maneuver Direction

The proximity of MRO's unphased position at the maximum particle fluency time to the desired safe location was the most important consideration in deciding which path to take to phase MRO. However, choosing which approach (speed up vs. slow down) has a profound effect on the atmospheric drag experienced by the spacecraft and hence, the ground track walk (GTW) error control needed for repeat cycles. During the planning of the maneuver strategy the timing offset from the safe location was estimated to be 25 minutes late. Since this phasing offset was less than a quarter of MRO's orbit, the more direct approach of speeding up was chosen (Figure 6). However, this required maneuvering in the anti-velocity direction to reduce the semi-major axis resulting in increased drag higher than that experienced during the PSO. This also took it to unprecedented higher levels of GTW errors reaching almost 240 km before being brought back via OTM-39, as seen in Figure 7b. In comparison, the GTW errors reached only about -90 km during Phoenix EDL support period (Figure 7a) and about 60 km during MSL EDL support period (Figure 7b). This higher GTW error was recognized as a temporary issue that would be corrected after the comet flyby and a secondary concern to the potential risk that the comet particles posed to the spacecraft.

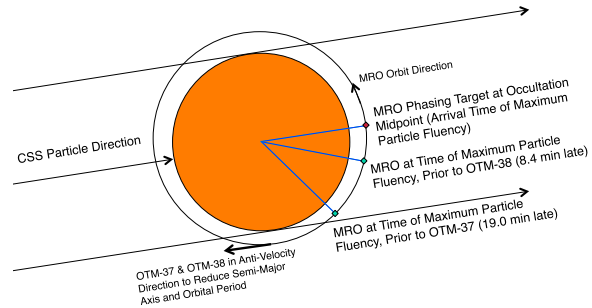
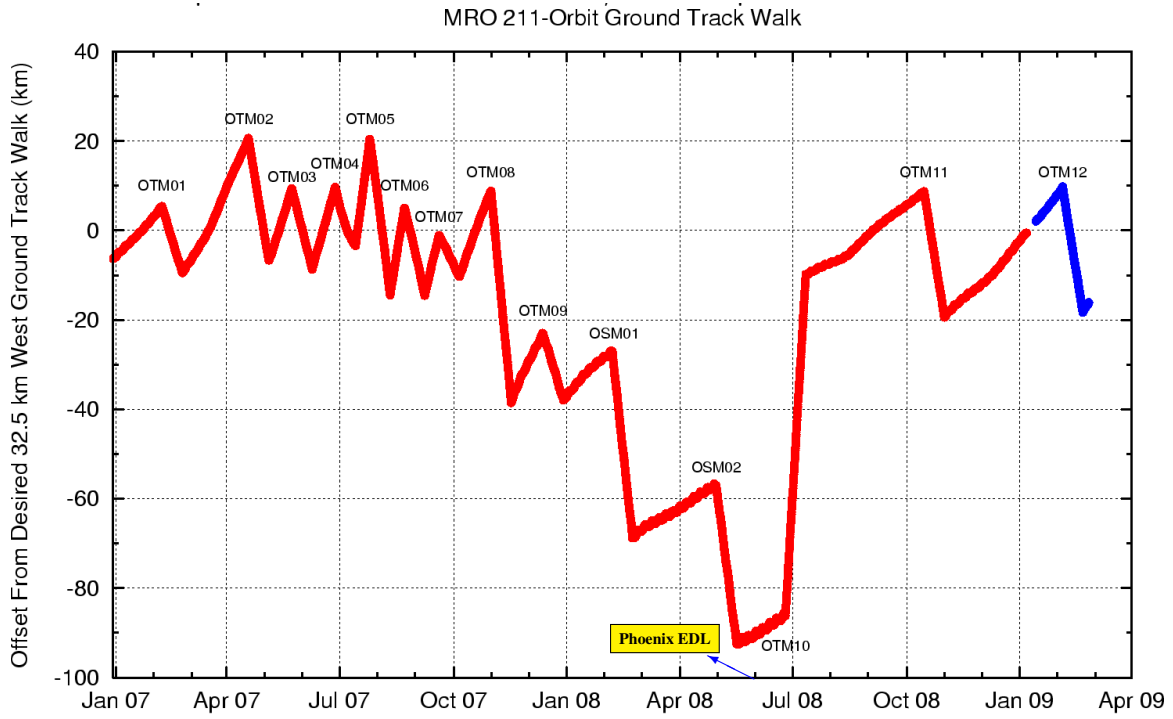
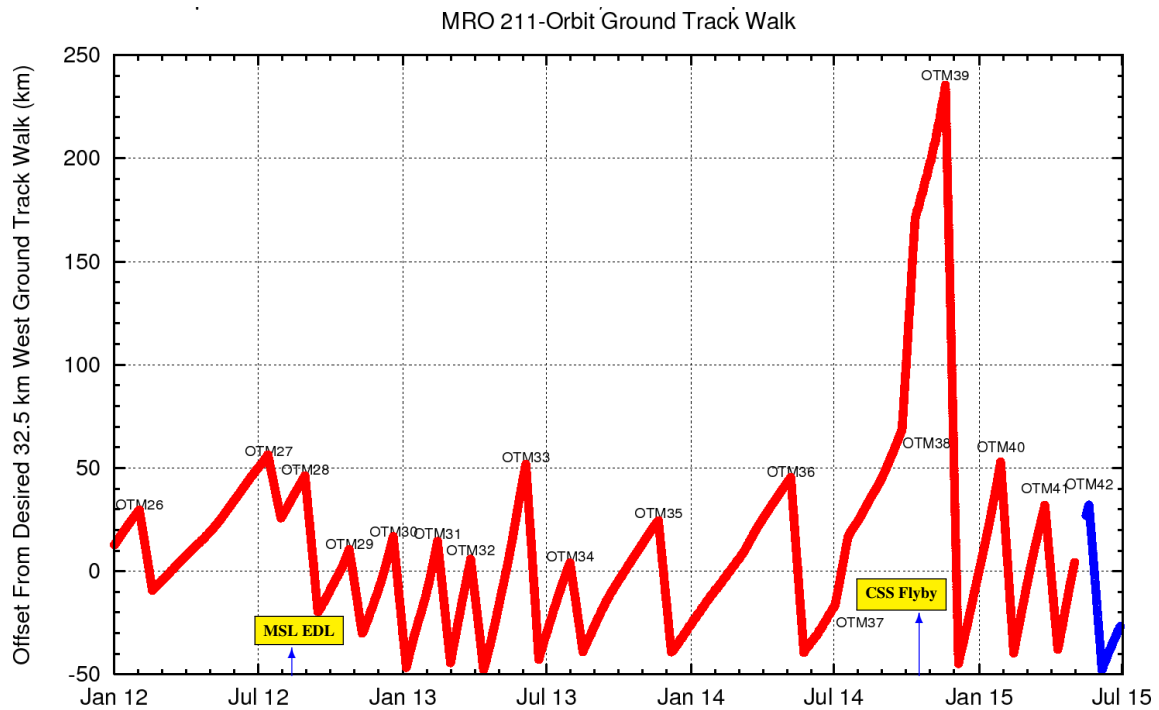


Figure 6: MRO Phasing Locations



(a) GTW Repeat Error from January 2007 – April 2009



(b) GTW Repeat Error from January 2012 – June 2015

Figure 7: MRO Orbit Ground Track Walk Repeat Error

Phasing Maneuvers (OTM-37 and OTM-38)

The MRO Navigation Team developed a maneuver plan for phasing MRO to the safe location at the time of the maximum particle fluency. The navigation timing uncertainties due to atmospheric drag and other orbital effects were among the chief considerations in the maneuver designs. Ultimately, a two-maneuver phasing approach was implemented for correcting the timing offset while minimizing the chance of overshooting the phasing target. As described in the previous section, both of these maneuvers would impart ΔV in the anti-velocity direction to speed up MRO so that it would reach the occultation midpoint at the designated hiding time.

OTM-37, the first of the two anti-velocity phasing maneuvers, was executed on July 2, 2014, three months prior to the comet flyby of Mars. Originally when the maneuver strategy was planned, the timing offset from the safe location was about 25 minutes late. However, at the time of the final maneuver design (one week prior to the OTM-37 execution) it had naturally drifted down to about 19 minutes late. Limiting the phasing correction to the navigation uncertainty level of 10.7 minutes (Figure 5), OTM-37 was designed to remove 9 minutes of phasing offset. This maneuver performed nominally with a slight overburn of 1.6%. Table 2 lists the estimated phasing errors at the safe location prior to and following each phasing maneuver, as well as the expected down-track timing uncertainties at each maneuver opportunity. Also shown in the table are the original and backup placements for the maneuvers that were cancelled.

Table 2: History of Phasing Offset from CSS Occultation Midpoint

	OTM-37	OTM-38 (original)	OTM-38	OTM-38 Backup	Comet Siding Spring Flyby
Maneuver Date	7/2/2014	8/27/2014	9/25/2014	10/1/2014	
Days Maneuver Prior to CSS Flyby	109	53	24	18	
Maneuver ΔV	0.0638 m/s	<i>cancelled</i>	0.2754 m/s	<i>cancelled</i>	
CSS Phasing Offset (Pre-Maneuver)	19.0 min (late)		8.4 min (late)		57 sec (early)
CSS Phasing Offset (Post-Maneuver)	10.0 min (late)		24 sec (early)		
Down-Track Timing Uncertainty (3σ)	10.7 min	3.0 min	48 sec	30 sec	
OD Data Cutoff (DCO) for Maneuver	6/23/2014	8/18/2014	9/16/2014	9/24/2014	
Days DCO Prior to CSS Flyby	118	62	35	25	

After OTM-37 a timing offset of about 10 minutes remained. This was planned to be corrected via OTM-38, the second planned phasing maneuver, on August 27, 2014. However, MRO went into safe mode just hours before the maneuver was scheduled to be performed. A favorable by-product of this unexpected safing event was that the phasing offset was reduced by about 28 seconds due to the increased drag caused by the spacecraft safe mode configuration. Instead, OTM-38 was executed one month later at the contingency location (September 25, 2014) to remove the remaining estimated offset of 8.4 minutes in orbit phase from the ideal safe spot. This maneuver also performed quite well (0.7% overburn) putting MRO early by only 24 seconds at its safe location. This was within the down-track timing uncertainty of 48 seconds at 35 days from the comet encounter date. From the MRO trajectory reconstruction following the comet flyby it was determined that MRO was early at the safe location by about 57 seconds. The increased attitude changes and atmospheric density variations, etc., had contributed to this offset. The difference between the right ascension of MRO's orbit and the safe location remained uncorrected at 32.40° (Figure 8).

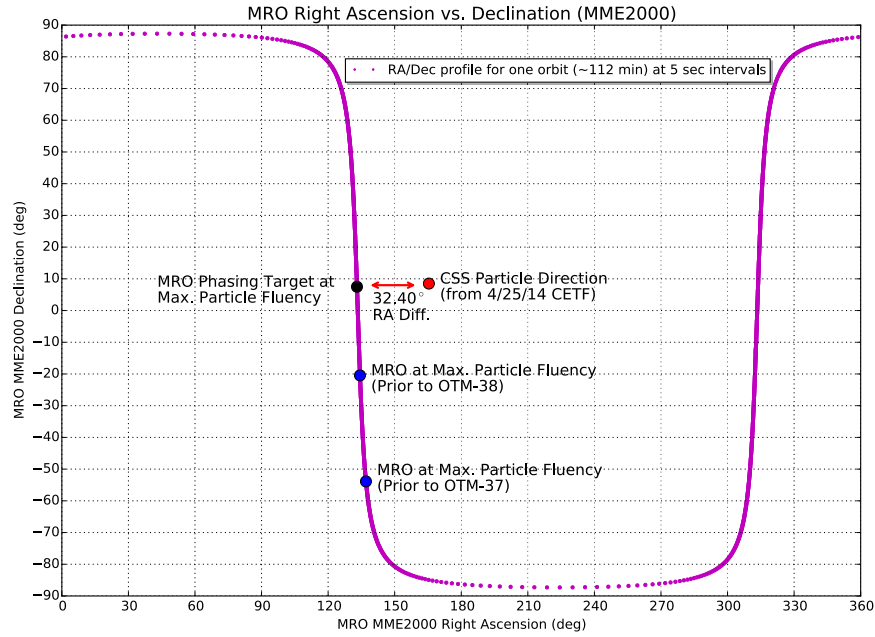


Figure 8: MRO Right Ascension/Declination at Maximum Particle Fluency Time

A summary of the design and reconstructed ΔV s of the two phasing maneuvers (OTM-37 and OTM-38) and the post-comet PSO recovery maneuvers (OTM-39 and OTM-40) discussed in a later section are given in Table 3, as well as the original and backup times of OTM-38. These four maneuvers were performed at different locations of the orbit – the two apses and the two equator crossings.

Table 3: MRO Maneuver History for CSS Phasing and PSO Recovery

Maneuver	Maneuver Epoch (UTC SCET)	Orbital Apsis/ Node	Data Source	ΔV (mm/s)		Right Ascension (deg)		Declination (deg)		Duration (sec)	
				Value	<i>err</i>	Value	<i>err</i>	Value	<i>err</i>	<i>Dur</i>	<i>err</i>
OTM-37	02-Jul-2014 14:12:08	Peri	Recon	64.9	1.1	299.26	0.46	−35.98	0.19	3.3	0.2
			Design	63.8		299.72		−35.79		3.1	
<i>OTM-38 (original)</i>	27-Aug-2014 13:46:56	Apo	 <i>Cancelled due to safe mode on 27-Aug-2014</i>								
			Design	155.3		156.42		35.57		7.2	
OTM-38	25-Sep-2014 13:21:09	Apo	Recon	277.3	1.9	174.46	0.21	31.68	0.04	12.2	0.3
			Design	275.4		174.25		31.64		12.5	
<i>OTM-38 (backup)</i>	01-Oct-2014 16:07:02	Peri	 <i>Contingency maneuver if OTM-38 on 25-Sep-2014 did not execute</i>								
			Design	344.9		357.29		−29.31		15.6	
<i>CSS Maximum Particle Fluency Time: 19-Oct-2014 20:07:00 UTC SCET</i>											
OTM-39	19-Nov-2014 13:21:44	DEq	Recon	3459.7	7.9	107.58	0.67	12.84	0.12	48.9	0.7
			Design	3451.8		108.25		12.72		49.6	
OTM-40	28-Jan-2015 13:36:57	AEq	Recon	434.2	0.7	315.21	0.20	55.48	0.08	18.8	0.6
			Design	433.5		315.41		55.40		19.4	

Phasing Offset Evolution History

The MRO Navigation Team started tracking the phasing offset after the completion of OTM-36 (May 7, 2014) and long before the first phasing maneuver OTM-37. The atmospheric density variations and other orbit determination inaccuracies were expected to affect the offset. As the orbit determination of MRO is done at least twice per week, it was possible to periodically assess the

current offset from the final desired target time. These were reported to the project via weekly Navigation Status Reports. Some of these periodic offset numbers and the associated navigation accuracies are given in Figure 9 and plotted with error bars. A negative sign in the offset indicates latency and a positive sign indicates being early. Thus prior to the second phasing maneuver OTM-38, MRO was expected to arrive late at the safe location, while after the maneuver it was to arrive slightly early.

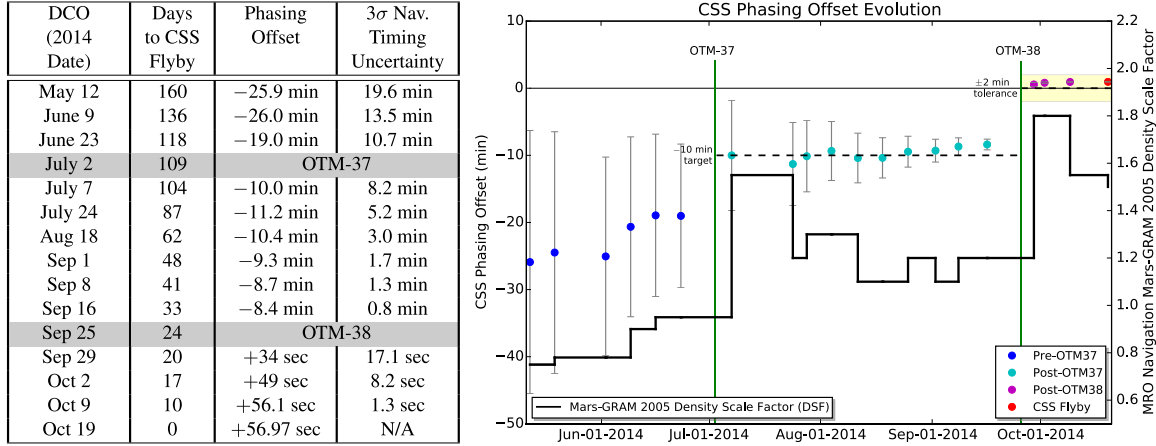


Figure 9: Evolution of the CSS Phasing Offset

COMET FLYBY OF MARS

Comet Siding Spring kept its date with planet Mars as expected on October 19, 2014. The time of closest approach was 18:27:31 UTC Spacecraft Event Time (SCET) at an estimated distance of $140,496.6 \pm 4.0 \text{ km}^{11}$ (based on JPL's comet orbit determination solution #102¹²).

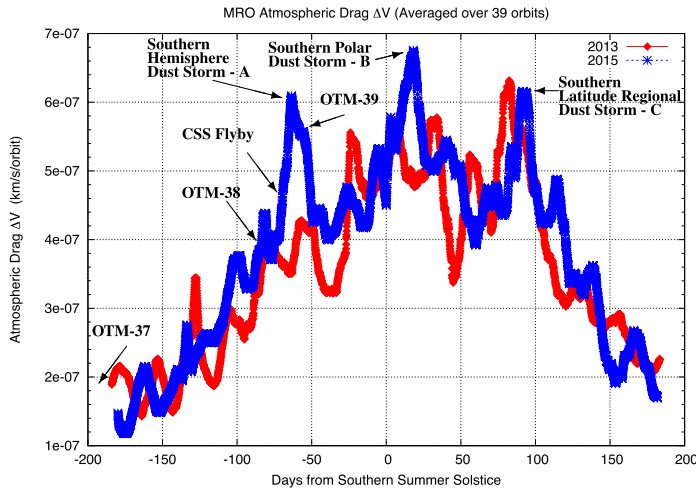


Figure 10: MRO Drag ΔV Per Orbit. Days from Southern Summer Solstice (Day 0: February 23, 2013 & January 11, 2015)

Coming on the heels of the comet flyby, the southern hemisphere dust storm-A on October 20 peaked on around November 5. Phasing to the safe location using anti-velocity maneuvers had decreased the mean semi-major axis by about 1-1.5 km, while the mean orbital period came down by about 3-3.5 seconds (Figure 11). After the comet flyby the steep decrease in drag ΔV from the pro-velocity maneuver OTM-39 that increased the semi-major axis is indicated in the figure. As shown in Figure 10 two more dust storms followed – a southern polar dust storm-B that started on December 17, 2014 and

a southern latitude regional dust storm-C from March 20, 2015. However, during this period the atmospheric density was near peak levels.

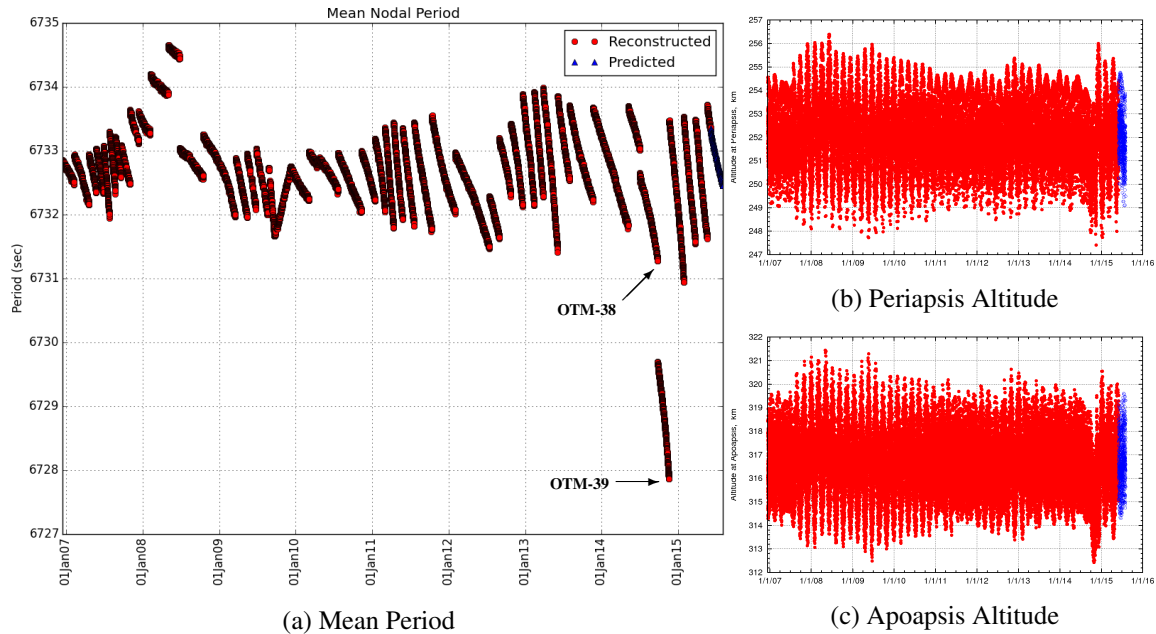


Figure 11: MRO Reconstructed Orbit Characteristics (January 2007 – July 2015)

The strategy to take advantage of the rare opportunity to image the comet and collect scientific data went as planned and the science teams achieved important findings (Reference 13). The Spacecraft Team at Lockheed Martin had done analyses on potential configuration changes of the spacecraft and its appendages (solar arrays, high gain antenna, etc.) for mitigating risk. However, based on the input from the CSS Risk Mitigation Team it was deemed unnecessary and hence was not implemented. Thus the preparations for the science observations were not compromised. Twelve days prior to the comet flyby, MRO took the first image of Siding Spring. Given the presence of non-gravitational forces on the comet, this early observation significantly improved the predicted trajectory of the comet around closest approach and ensured targeting of the high-resolution observations, but with small field-of-view. This and other comet ephemeris updates are maintained by JPL's Solar System Dynamics Group.¹² The Navigation Team provided viewing periods with and without atmospheric occultation effects to aid better observations of the comet. Table 4 shows a day of view periods centered at the comet closest approach.

Table 4: CSS View Periods (± 13 Hours from Closest Approach)

9/18/2014 MRO trajectory (with OTM-38 final design), 9/12/2014 CSS ephemeris (JPL orbit determination solution #67) CSS closest approach (CA) time: 19-OCT-2014 18:27:31 UTC SCET				
View Start (from CSS CA)	View Length	View Reduced by Atmosphere	View Start with 150 km Atmosphere (UTC SCET)	View End with 150 km Atmosphere (UTC SCET)
-12.9 hr	67.61 min	5.57 min	19-OCT-2014 05:30:51.6710	19-OCT-2014 06:38:28.0885
-11.1 hr	67.55 min	5.60 min	19-OCT-2014 07:22:49.7632	19-OCT-2014 08:30:22.5851
-9.2 hr	67.52 min	5.63 min	19-OCT-2014 09:14:41.6044	19-OCT-2014 10:22:13.0357
-7.3 hr	67.50 min	5.67 min	19-OCT-2014 11:06:26.2362	19-OCT-2014 12:13:56.3257
-5.5 hr	67.41 min	5.74 min	19-OCT-2014 12:57:54.5966	19-OCT-2014 14:05:19.1155
-3.6 hr	66.77 min	5.88 min	19-OCT-2014 14:48:44.7184	19-OCT-2014 15:55:30.7499
-1.8 hr	60.86 min	5.76 min	19-OCT-2014 16:37:02.3383	19-OCT-2014 17:37:54.2095
-0.3 hr	50.07 min	4.23 min	19-OCT-2014 18:10:15.1028	19-OCT-2014 19:02:19.0274
+1.3 hr	63.02 min	4.54 min	19-OCT-2014 19:42:31.6558	19-OCT-2014 20:45:32.7896
+3.1 hr	65.66 min	4.97 min	19-OCT-2014 21:30:27.8909	19-OCT-2014 22:36:07.2628
+4.9 hr	66.41 min	5.13 min	19-OCT-2014 23:21:12.2483	20-OCT-2014 00:27:36.7427
+6.8 hr	66.78 min	5.19 min	20-OCT-2014 01:12:38.9051	20-OCT-2014 02:19:25.7151
+8.6 hr	66.99 min	5.22 min	20-OCT-2014 03:04:24.2121	20-OCT-2014 04:11:23.4327
+10.5 hr	67.10 min	5.25 min	20-OCT-2014 04:56:19.9389	20-OCT-2014 06:03:25.9670
+12.4 hr	67.13 min	5.27 min	20-OCT-2014 06:48:21.6216	20-OCT-2014 07:55:29.2364

Science Returns

The scientific findings by MRO resulted in the improved knowledge of the comet's rotation period. The MRO images led to the resolution of the lit part of the comet nucleus as shown in Figure 12. The morphology and composition of the coma were better understood. Four jets were detected on the comet of which three were on the Sun-side at the time of the observation. No significant temperature or dust/ice aerosol was detected by MRO in the lower or middle atmosphere of Mars due to the comet flyby. However, an enhanced total electron count was detected 2.5 hours after the closest approach time.¹³ These scientific results were included in the MRO Science Team's presentation to the American Geophysical Union in December 2014 (Reference 6).

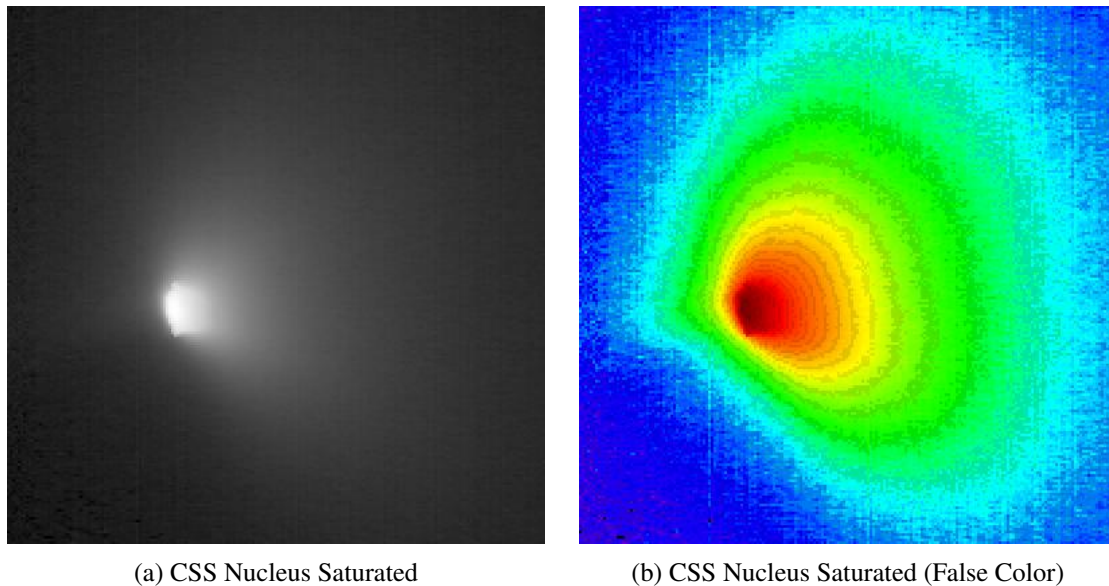


Figure 12: Close Approach Image of Comet Siding Spring Taken by HiRISE Camera. 18:24 UTC SCET at a range of 139,000 km, 28×28 km field-of-view. Source: Alan Delamere.

POST-COMET RETURN TO MRO PRIMARY SCIENCE ORBIT

Reducing the orbit semi-major axis with anti-velocity phasing maneuvers OTM-37 and OTM-38 had moved the GTW error well outside the mission requirements for approximately three months. Also by this time due to natural drift the Local Mean Solar Time (LMST) was inching towards 3:10 PM. The nominal LMST needed to support PSO is 3:00 PM with ± 15 -minute bounds. Thus the return to the PSO was scheduled using maneuver OTM-39 for November 19, 2014, a month after the comet encounter. This utilized a pro-velocity in-plane component to re-establish the GTW control and an out-of-plane (inclination change) component to send the LMST towards 3:00 PM. This turned out to be the biggest maneuver performed during the PSO with a ΔV of about 3.45 m/s. It performed well within expectations (0.2% overburn). Finally, OTM-40 was executed on January 28, 2015 with an in-plane ΔV of 0.43 m/s at an ascending equator crossing to control the GTW error. This maneuver marked the first time a purely GTW maintenance maneuver was performed at a non-apsis location. The MRO's frozen orbit had the periapsis fixed at the south pole with the ascending node at 3:00 PM. The frozen condition was tracked via the mean $e - w$ space. The frozen condition did not get adversely affected during the performance of OTM-37 at the periapsis.

However, due to the OTM-38 execution at the apoapsis and to a greater degree by OTM-39 with a significant ΔV done at the descending equator crossing, the frozen condition had greatly eroded. The latter maneuver also included an in-plane component of about 1 m/s for GTW control. Finally, OTM-40 at the ascending equator crossing helped partially recover the frozen orbit condition by essentially countering the effects of OTM-39. The above described erosion and retrieval of MRO's frozen condition are shown in Figure 13.

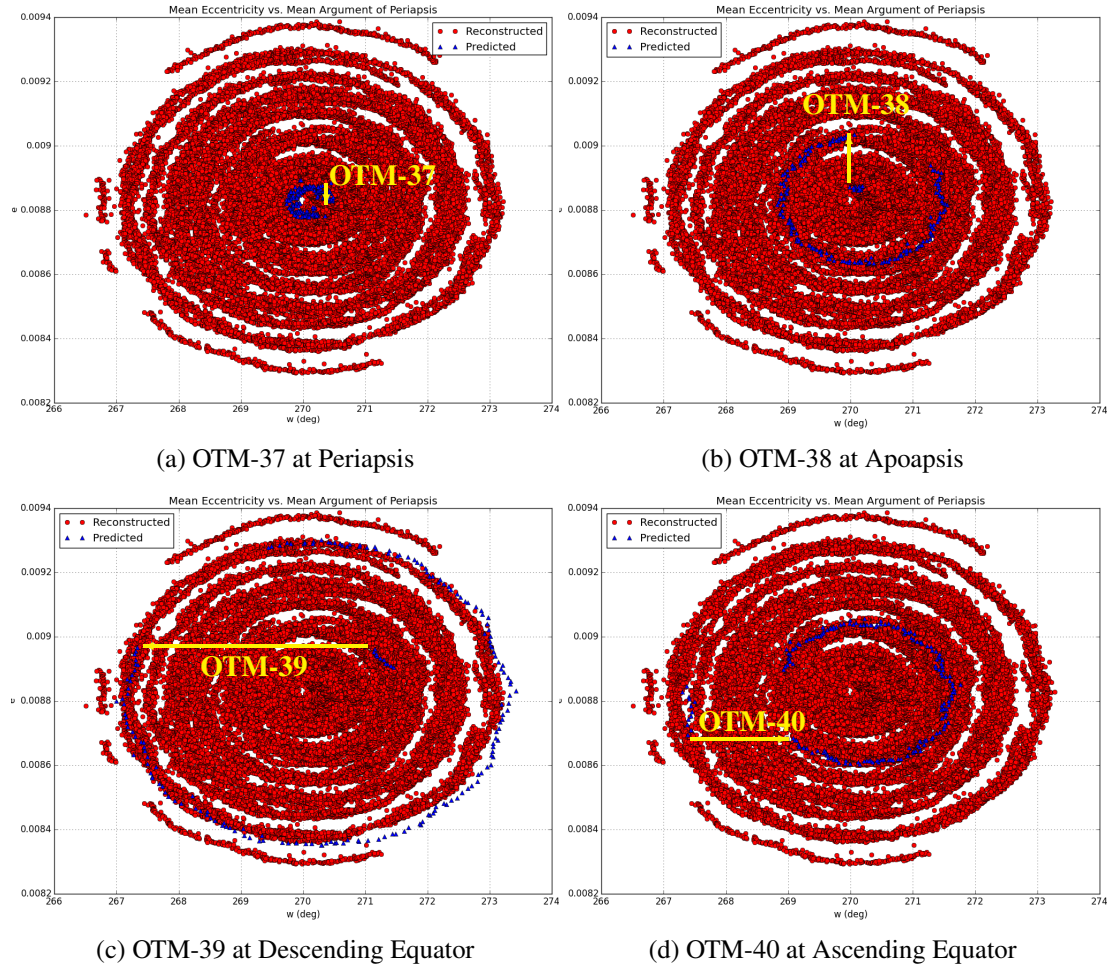


Figure 13: OTMs 37–40 Effects to MRO PSO Frozen Condition (Mean $e - w$)

CONCLUSION

As expected Comet Siding Spring arrived on time and left without any significant effect on the martian environment. The MRO navigation plan to mitigate risk from the comet particles was successfully implemented with the executions of two maneuvers prior to the comet flyby. Thus MRO had arrived at the minimum risk location early only by about 57 seconds, well within the phasing requirement. This paved the way for a productive observation campaign of the comet and study of its effect on the martian environment. MRO has now transitioned back to its prime duties of supporting science observations in the PSO and relay support to the martian rovers. It is now preparing for the EDL support of the InSight Mission in 2016. The success of the comet risk mitigation and science observation campaigns may be attributed to the tremendous synergy between

various teams including CSS Risk Mitigation, Spacecraft Engineering, Navigation, Science, Solar System Dynamics, Mission Planning, Payload Operations Support, Ground Data Systems, and Deep Space Network.

APPENDIX: COMET ENCOUNTER TARGET FILE

The Comet Encounter Target File (CETF) provided the particle grain direction's right ascension and declination in the Mars Mean Equator of J2000 (MME2000) inertial reference frame, as well as the expected arrival time of the particle fluence center. A tolerance of ± 2 minutes was also given for the target time for the Navigation Team to achieve. It turned out that this CETF did not require an update and hence the safe target was based on the the CETF issued on April 25, 2014.

```
Comet Encounter Target File (CETF)

Generated April 25, 2014 by R. Lock

*****
Hiding Zone center location (Mars Mean Equator of J2000 reference frame)

Right Ascension:      165.4 deg
Declination:          8.5 deg
Time of particle fluence center:  2014 Oct 19 20:07 (UTC-SCET)
Tolerance of time estimate:      plus/minus 2 minutes

*****

Note: the time of particle fluence center is the time specified by the Mars program as 98 minutes after the
closest approach.

Sources:

Comet C/2013 A1 Siding Spring

Solution #46
SPK:  ftp://ssd.jpl.nasa.gov/pub/xfr/c2013a1_s46_merged_DE431.bsp      (Binary SPK format)

C2013A1_delivery_memoirandum_2014-3-31.doc

CSS_project_brief_2014_04_14c.ppt
```

Figure 14: Comet Encounter Target File (CETF)

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